

9.11 Determining Surface Area Given Dimension Changes

Lesson Introduction

 Benchmark	MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the areas of two-dimensional figures and the surface area or volume of three-dimensional figures.		 Learning Targets	- I can find the area of a figure when one or both dimensions are changed. - I can find the surface area of a three-dimensional figure when one or both dimensions are changed.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- How can you find the area of a figure when one or both dimensions are changed? - How do you find the surface area of a three-dimensional figure when one or both dimensions are changed? - What effect(s) does changing the dimensions have on the area or surface area of a figure?				
 Lesson Narrative	The lesson guides students to notice the impacts that changing dimensions have on area and surface area and provides several real-world scenarios that students work through and discuss with classmates. The lesson builds on previous lessons on area, volume, and surface area to discover the impact that dimension changes have on the area, volume, and surface area.				
 Component Summary	9.11.1 Warm-Up (~5 min.)	Determining the area of a trapezoid and the coordinates of a dilated image (<i>SE p. 112</i>)	9.11.3 Guided Practice (10 min.)	Finding area and surface area when dimensions are changed (<i>SE p. 115</i>)	
	9.11.2 Exploration (15-20 min.)	Evaluating how changing dimensions impacts area and cost of a home project (<i>SE p. 113</i>)	9.11.4 Your Turn! (5-10 min.)	Determining how the area changes when dimensions are changed (<i>SE p. 116</i>)	
	9.11.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 116</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Scale factor - Dilation		(Optional) Chart Paper for Students to Record Results (Optional) Colored Pens/Pencils (Optional) Graphing Utility		If necessary, find a GeoGebra applet using the phrase “Scale Factor and Area” and “Scale Factor and Volume.”

 Teacher Tips	Common Misconceptions	Things to Consider
	- On question 7 in 9.11.2 Exploration, students may forget to subtract the part of the surface that is water when calculating the surface area (see notes in lesson guide). - Students may not apply the correct formulas for area, volume, and/or surface area.	Consider having students record their results in groups using chart paper or other forms of recording data.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect This lesson provides an opportunity for students to connect and compare their learning to a real-world scenario. Provide opportunities for students to reflect on their learning throughout the lesson.	MA.K12.MTR.7.1: Real-World Contexts Lesson 11 is written purposefully using real-world concepts that have students connect mathematics to everyday experiences. Allow students to work through the questions in the lesson and use various methods to solve.
 Differentiate Instruction	Consider making the 9.11.2 Collaboration, 9.11.3 Guided Practice, and 9.11.4 Your Turn! into separate stations that students can work through. - Station 1: Collaboration (questions 1-3) - Station 2: Collaboration (questions 4-6) - Station 3: Collaboration (question 7-9); be sure that students have completed Stations 1 and 2 first. Station 3 may also be a teacher-centered station. - Station 4: Guided Practice - Station 5: Your Turn!	
Lesson Notes		

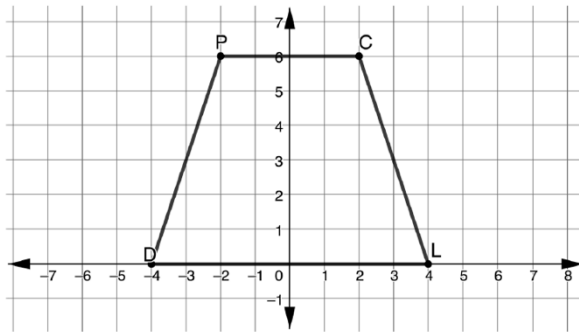
9.11.1 Warm-Up: Bell Work

Component Overview: Students find the area of the given trapezoid and then perform a dilation with a given scale factor. This ties into Lesson 10.



9.11.1 Warm-Up: Bell Work

1. Trapezoid $DPCL$ is shown in the coordinate plane.



- a. Find the area of $DPCL$.

$$A = \frac{1}{2}(8 + 4)(6)$$

$$A = 36 \text{ unit}^2$$

- b. Find the coordinates of the image of $MTRD$ given the scale factor in the table. Use the preimage coordinates to find the coordinates of the image. The dilation is centered at $(0,0)$.

Point	Preimage	Dilation $k = 6$ ($D'P'C'L'$)
D	$(-4,0)$	$(-24,0)$
P	$(-2,6)$	$(-12,36)$
C	$(2,6)$	$(12,36)$
L	$(4,0)$	$(24,0)$



Can you find the area of the trapezoid by dividing into triangles and rectangles?



Consider allowing students to use either the formula for the area of a trapezoid or divide into triangles and a rectangle.

9.11.2 Exploration: Ryan's Renovations

Component Overview: 9.11.2 Exploration guides students through scenarios that would be used in home renovations, thus providing a real-world context. Students find the area of a figure and apply new dimensions through changing the sizes of a kitchen and a swimming pool.



9.11.2 Exploration: Ryan's Renovations

Ryan is building a new house and wants to make the size of the kitchen larger. The current kitchen is 24 feet wide and 10 feet long. Ryan wants to make the new kitchen 20 feet long but leave the width the same.

- Find the area of the original kitchen and the area of the new kitchen.

Area of the Original Kitchen	Area of the New Kitchen
$A = 10 \cdot 24$ $A = 240 \text{ ft}^2$	$A = 20 \cdot 24$ $A = 480 \text{ ft}^2$

- How do the areas compare?
The area of the new kitchen is twice the area of the original kitchen.

Ryan also wants to install a pool for his new house. The construction company gives Ryan three shape options for the new pool. Each pool's dimensions are given in feet (ft).

Pool 1	Pool 2	Pool 3
Cylindrical	Rectangular Prism	Cube
diameter = 16 ft height = 12 ft	length = 30 ft width = 20 ft height = 12 ft	Each side = 15 ft

- Find the surface area of each of the shapes for the three pool options. Round to the nearest thousandth.

Pool 1	Pool 2	Pool 3
$SA = 2B + Ch$ $SA = 128\pi + 192\pi$ $SA = 320\pi$ $SA \approx 1,005.310 \text{ ft}^2$	$SA = 2B + Ph$ $SA = 1,200 + 1,200$ $SA = 2,400 \text{ ft}^2$	$SA = 6s^2$ $SA = 6(225)$ $SA = 1,350 \text{ ft}^2$



Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions and which students help clarify.



Consider using 9.11.2 Exploration as three separate stations (see Differentiate Instruction notes on page 2 of the lesson guide).



The collaboration gives an opportunity to fit in MA.K12.MTR.7.1: Real-World Contexts, where students are applying mathematics to real-world contexts.



On Pool 1, have students divide the diameter by 2 to find the radius.

The part of the surface area that is water is included in these calculations. That part of the surface area will be subtracted later in the lesson.

9.11.2 Exploration: Ryan's Renovations (continued)

The construction company is giving Ryan more options for building a swimming pool. The three options are listed in the table.

Pool 4	Pool 5	Pool 6
Cylindrical	Rectangular Prism	Cube
diameter = 8 ft height = 6 ft	length = 15 ft width = 10 ft height = 6 ft	Each side = 7.5 ft

4. Describe how the dimensions of pair of pools are related.

Answers will vary.

Pair	Description
Pool 1 and Pool 4	<i>The dimensions of Pool 1 are twice as large as the dimensions of Pool 4.</i>
Pool 2 and Pool 5	<i>The dimensions of Pool 2 are twice as large as the dimensions of Pool 5.</i>
Pool 3 and Pool 6	<i>The dimensions of Pool 3 are twice as large as the dimensions of Pool 6.</i>

5. Find the surface area of each of the shapes for the three new options for a pool. Round to the nearest thousandth.

Pool 4	Pool 5	Pool 6
$SA = 2B + Ch$ $SA = 32\pi + 48\pi$ $SA = 80\pi$ $SA \approx 251.327 \text{ ft}^2$	$SA = 2B + Ph$ $SA = 300 + 300$ $SA = 600 \text{ ft}^2$	$SA = 6s^2$ $SA = 6(56.25)$ $SA = 337.5 \text{ ft}^2$



Consider opportunities to have students display their work in the classroom when working through the collaboration. Chart or regular paper can be used for each pool.



How did the areas of Pools 4, 5, and 6 differ from Pools 1, 2, and 3? (The surface areas of Pools 4, 5, and 6 are $\frac{1}{4}$ of the surface areas of Pools 1, 2, and 3.)



The part of the surface area that is water is included in these calculations. That part of the surface area will be subtracted later in the lesson.

9.11.2 Exploration: Ryan's Renovations (continued)

6. Record the surface area of all pool shapes in the table below.

Pool	Surface Area
1	$SA \approx 1,005.31 \text{ ft}^2$
2	$SA = 2,400 \text{ ft}^2$
3	$SA = 1,350 \text{ ft}^2$
4	$SA \approx 251.327 \text{ ft}^2$
5	$SA = 600 \text{ ft}^2$
6	$SA = 337.5 \text{ ft}^2$

The cost of building the swimming pool is dependent upon the amount of material. Since the top of each of the shapes would be water, determine the surface area of each pool that could be used to determine the cost.

7. Complete the table.

Pool	Surface Area for Material
1	$SA = 804.248 \text{ ft}^2$
2	$SA = 1,800 \text{ ft}^2$
3	$SA = 1,125 \text{ ft}^2$
4	$SA = 201.062 \text{ ft}^2$
5	$SA = 450 \text{ ft}^2$
6	$SA = 281.25 \text{ ft}^2$

8. If Ryan wants the least expensive pool, which one should he purchase?

Ryan should purchase Pool 4 because it has the smallest surface area.

9. Which pool would you purchase? State your reasoning.

Answers will vary.

Pool 2 because it will give more space in which to swim.



Students must subtract the surface that is the water in question 7. Monitor students' work to show that this is reflected in their final answers.



Consider providing students an opportunity to discuss their personal choices and reasons for which pool they would purchase.

9.11.3 Guided Practice: Ryan's House

Component Overview: 9.11.3 Guided Practice gives a scenario about the installation of a shower and a bathtub. Students find the surface area to determine whether the shower or tub should be installed.



9.11.3 Guided Practice: Ryan's House

Ryan is building a bathroom and needs to install either a bathtub or a shower. The options for each bathroom size are given.

Shower 1	Tub 1
length = 4 ft width = 6 ft height = 12 ft	length = 4 ft width = 8 ft height = 10 ft

- In order to determine the amount of drywall, Ryan needs to find the surface area of each of the room options.

Shower 1	Tub 1
$SA = B + Ph$ $SA = 24 + 240$ $SA = 264 \text{ ft}^2$	<i>Calculating SA without the top.</i> $SA = B + Ph$ $SA = 32 + 240$ $SA = 272 \text{ ft}^2$

- Ryan determines that the dimensions of the shower and tub are not up to code. The updated dimensions are shown. Use the new dimensions to determine the surface area of each room.

Shower 2	Tub 2
length = 6 ft width = 9 ft height = 12 ft	length = 6 ft width = 12 ft height = 15 ft
Surface Area	Surface Area
$SA = B + Ph$ $SA = 54 + 360$ $SA = 414 \text{ ft}^2$	$SA = B + Ph$ $SA = 72 + 540$ $SA = 612 \text{ ft}^2$



Continue highlighting MA.K12.MTR.7.1: Real-World Contexts in 9.11.3 Guided Practice.



Make sure that students do not find the surface area of the top of the tub.

9.11.3 Guided Practice: Ryan's House (continued)

Two of Shower 2's dimensions were changed by a factor of 1.5, whereas all of Tub 2's dimensions were changed by a factor of 1.5. Discuss with your partner on how to use the factor 1.5 and the number of changes in dimensions to find the new surface area from the original surface area.

3. Complete the statements.

1.568 $\times$ surface area of Shower 1 = surface area of Shower 2

2.25 $\times$ surface area of Tub 1 = surface area of Tub 2



Students are not required to write a summary of their discussions. Listen as students discuss their thinking in response to this prompt. Student thinking should highlight that the scale factor is related to the area if the scale factor is applied to all dimensions of the shape but not just some dimensions.



Question 3 is algebraic equations that are solved by substituting in the values from the tables above.

9.11.4 Your Turn!

Component Overview: Students work through a scenario involving a living room and are asked to find the area and analyze the effect of a dimension change with a given scale factor.



9.11.4 Your Turn!

The living room of Ryan's house is 12 feet long and 18 feet wide.

1. The room is a rectangular shape. Find the area of Ryan's living room.

$$A = 216 \text{ ft}^2$$

2. How would the area of the living room change if Ryan decided to enlarge the dimensions by 25%?

Old Dimensions	New Dimensions
Length = 12 ft	Length = 15 ft
Width = 18 ft	Width = 22.5 ft

New area = 337.5 ft². This is an enlargement.

3. What is the scale factor for a dimension increase of 25%?

The scale factor is 1.25.

4. How did the area of the living room change with the new dimensions?

$$1.25^2 = 1.5625$$

The area of the living room increased by 56.25%



Continue highlighting MA.K12.MTR.7.1: Real-World Contexts in 9.11.4 Your Turn!



On question 3, students may forget to add 100% to the 25% to find 1.25 as the scale factor.



For students who finish early, challenge them to create dimensions for another room and how the area would change with new dimensions based on a scale factor for another room in Ryan's house.

9.11.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



9.11.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.